

Werner Bisschoff

EMBEDDED SYSTEMS & REAL-TIME SOFTWARE ENGINEER

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Dear [Hiring Manager Title / Engineering Lead Name],

About Me

I am writing to express my interest in the Senior Systems Engineering position at [Company]. I specialize in engineering deterministic, real-time software architectures (C/C++, FreeRTOS, Elixir/OTP) and secure, spec-driven communication protocols. Throughout my career, my focus has been on eliminating runtime faults and managing high-concurrency systems at the edge and data boundaries.

Why Me?

Recently, as Principal Systems Architect for the open-source Divergent Tabletop ecosystem, I engineered a high-concurrency real-time wiki engine using Elixir and Rust NIFs (y_det). By implementing a dynamic supervision tree and an actor-based memory buffering architecture, I achieved 99.9% fault isolation and reduced per-session memory overhead by 60%. My background also includes engineering low-level NFC communication stacks (ISO 14443-4 APDUs) for display infrastructure at FARO Africa and modeling dual-core ESP32 finite state machines at Ingenics Digital GmbH.

How I Work

I approach software through the lens of Spec-Driven Development (SDD) to ensure code correctness and maintain clear, robust architectures. I am looking forward to bringing this systems-level discipline to [Company]'s engineering team.

Best Regards,

Werner Bisschoff